



SCI_BUILD_A2_W05 · BUILD Science teacher note

Staff-facing; review only. Verbatim SoW objective: Group natural and human-made materials; record results in a table (ICT).

Source: BUILD scheme of work, Autumn 2 week 5. Pupil goal: I am learning to sort things from nature and things people make.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can sort one raw item., I can sort one made item., I can add a row to a table.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	Name one use of a rock.	Slate can cover roofs, where suitable.	BUILD scheme of work, Autumn 2 week 4
2-4 weeks	What does a water test check?	Whether water passes through the sample.	BUILD scheme of work, Autumn 2 week 3
last term	What do people and animals get from food?	Nutrition.	BUILD scheme of work, Autumn 1 week 5

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Is a finished brick raw clay, or an item made by people? Answer: It is made by people from clay. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
natural	Tier 2	from nature, not made into a new item	A raw rock is natural.
human-made	Tier 3	made or changed by people	People make a brick from clay.
material	Tier 3	what a thing is made of	The brick has clay in it.
table	Tier 2	rows and columns for facts	This row says brick: made.

Faded example, tiered task and stretch

I do full: "This raw rock is from nature. I write raw rock in the Natural row." We do part-done: "People shape and fire clay to make a brick. I write brick in the row." Removed step: Choose Human-made as the group for the finished brick. Independent: no filled step. Next fade: Give the raw and finished item with a blank table; pupil fills both rows.

We do F: Hold a raw rock. Point to its group. M: Sort four safe items. Tell your rule. H: Where do raw clay and finished brick go? Why? Independent F: Put one raw and one made item in groups. Fill one row. M: Sort four items. Add two rows to a table. H: Why is clay natural but a brick made? Go deeper: An object has two materials. How can your table show both? Named harder stretch — Mixed item judge: Choose a made item with two raw materials. Plan a fair row.



The kit route is hands-on or sensory: clean raw rock, brick card, leaf card, safe bottle picture. Two on-screen interactions are card sort and Which table row is right?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Hinge and decision

Question: People shape sand into a glass bottle. What is the finished bottle?

Option	Correct?	Diagnosis
A raw, natural sand sample.	no	Confuses the source material with the finished object.
A raw and finished rock.	no	Calls a finished item raw at the same time.
A human-made glass item.	yes	Names the finished object as made by people.
A thing outside all groups.	no	Misses the natural or human-made grouping.

Rule: About 3 in 4 or more right → move on; otherwise re-model This raw rock is from nature. I write raw rock in the Natural row. Ask the parallel hinge. No scores or answers saved. Parallel hinge: People cut wood and assemble a chair. Is the finished chair raw wood or a human-made item? A human-made item.

Misconceptions:

Misconception	Spot	Correct
A made thing is natural whenever its raw material is natural.	Calls finished brick raw clay.	Put source clay and finished brick in separate rows.
A table is only a picture of groups.	Cannot name row and column information.	Point to the item column and group column, then make one row.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ? At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ? R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Give a three-step visual card for natural and made materials; remove its last cue after a correct try.
Communication & Interaction	explicit instruction	Point to the real kit before a short spoken, signed or AAC response; wait for a turn.
SEMH	cognitive/metacognitive strategies	Offer a private first-then cue and planned pause before the hinge; re-enter with one step.
Sensory/Physical	technology	Use a stable tray, large cards or an accessible input; no forced touch.



Broad area	EEF five-a-day strategy	Specific support
EAL (language, not SEN)	flexible grouping	Rehearse a full M or H idea in any language before writing; keep the science challenge.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the object or model. → Ask: What will you check next? → Model one step and give the next turn back. → Finish one example together, then let the pupil try. No physical prompt. Stage scripts: I do “Watch my first step. What do you see?”; We do “What step is missing? Show it with the kit.”; Independent “What can you do first on your own?”. At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Point to the raw rock. → The unmade rock sample; What is a table? → Rows and columns for facts.

- M: Is a finished brick made by people? → Yes. People make it from raw clay; What is a material? → What a thing is made of.

- H: How can clay be natural but brick made? → Clay is raw material; people shape and fire the brick; How can a table show two raw materials? → Add two material entries for one made item.

Regulation: Agree a quiet pause and supervised place. Give one short cue. Re-enter at the last completed step. No evidence while distressed.

lesson record: BUILD_SCI_A2_W05 via staff device or projector only. Gatsby 4: A materials sorter records which objects were made by people.

Reading gate: BUILD below age 8; rough ARI grade 1.57 on 619 pupil-visible words (adult guidance and navigation excluded); heuristic age guide 7.57 years, not a formal reading-age test. This is a rough screen, not an individual assessment.

Prepared model handling · practical safety

Kit: Clean blunt raw rock, picture cards of made items, tray

Method: Hold the checked rock, sort it with the made-item cards, then fill a table row.

Pupil Risk: A raw rock could have a rough edge or loose dust.

Pupil Safe: Use only a clean blunt checked rock on a tray. Do not taste or scrape it; wash hands after.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-BUILD-A2-W05

SoW reference: BUILD scheme of work, Autumn 2 week 5

Exact SoW outcome: Group natural and human-made materials; record results in a table (ICT).

Evidence (scheme of work): Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit



assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-BUILD-A2-W05. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Hold a safe raw rock. Look at a brick card. Staff: Offer a safe kit choice and short first-then cue.

Starter (3 min): Think back. Say, sign or point. Staff: Ask three low-stakes prior questions; answers in note.

I Do (5 min): Watch a row go in a table. Staff: Pre-teach words; show the full model; ask the show-me check.

We Do (7 min): Sort safe samples and cards. Staff: Hand over the missing step; offer F or H route. Reset once.

We Do 2 (5 min): Choose one answer. All of us show it. Staff: Take all pupil responses at once; use decision rule and parallel hinge.

Independent (8 min): Make a row for raw and made items. Staff: Choose one adult for supported pupils, one for group; swap across the week.

Review (4 min): Check one row. Hear one next step. Staff: Give VF and one NS(+); invite one self-edit. Reset once.

Exit (3 min): Show your row to an adult. Staff: Hear feedback and next step, choose next start, then one-minute reset.

Planned adaptation — cognition learning: Give a three-step visual card for natural and made materials; remove its last cue after a correct try.

Planned adaptation — communication interaction: Point to the real kit before a short spoken, signed or AAC response; wait for a turn.

Planned adaptation — semh: Offer a private first-then cue and planned pause before the hinge; re-enter with one step.

Planned adaptation — sensory physical: Use a stable tray, large cards or an accessible input; no forced touch.

Planned adaptation — eal: Rehearse a full M or H idea in any language before writing; keep the science challenge.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: explicit instruction; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: flexible grouping

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Make a row for raw and made items. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W06. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W06: Plan and carry out a simple fair test on rocks.



If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Choose Human-made as the group for the finished brick. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "People shape sand into a glass bottle. What is the finished bottle?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Hold a safe raw rock. Look at a brick card. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Make a row for raw and made items. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use natural, human-made, material, table with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.